INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

COMPACTOR, HIGH SPEED, SELF-PROPELLED, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high speed, self-propelled, diesel powered, tamping foot, embankment compactor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-V-173	- Varnish, Moisture-and Fungus-Resistant, (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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MIL-T-704	- Treatment and Painting of Material.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-R-3075	- Rollers, Motorized, Road, Diesel-or Gasoline-Engine-Driven, Packaging of.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Gun, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-T-62314	- Transmission, Power Shift, Commercial Modified.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property .
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35844	- Plug Machine Thread, Magnetic; Drain.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Functional Fluids, Preservatives, and Specialty Products for Use in Ground Equipment.
MIL-HDBK-157	- Transportability Criteria.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

US ARMY TANK-AUTOMOTIVE COMMAND (TACOM)

Report No. CR-82-588-003 - STE/ICE Design Guide for Vehicle Diagnostic Connector Assembly.

DRAWINGS

TACOM

12258955 - Vehicle Test Card.

(Copies of the TACOM report and drawing required by contractors in connection with specific acquisition functions should be obtained from the USA Tank Automotive Command, ATTN: AMSTA-MD, Warren, MI 48397-5000.)

ME

13229E8014 - Bracket Assembly Rifle Stowage.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FMT, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1557 - Moisture-Density Relations of Soils and Soil-Aggregate Mixtures using 10-lb. (4.54-KG) Rammer and 18-in. (457-MM) Drop.
D 2487 - Classification of Soils for Engineering Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

14.3 - Specification for Welding Earthmoving and Construction Equipment.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P. O. Box 351040, Miami, FL 33135.)

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ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street NW, Suite 7702, Washington, DC 20001.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J154 - Operator Enclosures Human Factor Design Considerations.
J185 - Access Systems for Off-Road Machines.
J209 - Instrument Face Design and Location for Construction and Industrial Equipment.
J276 - Steering Frame Lock for Articulated Loaders and Tractors.
J297 - Operator Controls on Industrial Equipment.
J386 - Operators Restraint Systems for Off-Road Work Machines.
J514 - Hydraulic Tube Fittings.
J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30 - 1000 MHz).
J753 - Lubrication Chart - Construction and Industrial Machinery.
J833 - USA Human Physical Dimensions.
J898 - Control Locations for Off-Road Work Machines.
J899 - Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines.
J1040 - Performance Criteria for Rollover Protective Structures (ROPS) for Construction, Earthmoving, Forestry and Mining Machines.
J1362 - Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The compactor shall be a self-propelled, diesel powered, tamping machine for high speed embankment compaction. The compactor shall be a commercial model consisting of articulated steering, a hydraulically controlled strike-off dozer blade, and all wheels shall have tamping foot or segmented impact pads with adjustable wheel cleaners. The compactor shall be complete with all components furnished as standard equipment that are customarily offered with a compactor of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the compactor to function reliably and efficiently in a sustained operation. The compactor shall conform to all EPA and OSHA legal requirements for US domestic equipment in effect at the date of manufacture.

3.1.1 Dimensions. The compactor shall conform to the following maximum dimensions:

Length - 225 inches (without blade.)
Width - 107 inches (without blade.)
Height - 102 inches (exhaust stack and ROPS removed.)

3.1.2 Performance. The compactor shall be capable of compacting at least 1500 compacted cubic yards per hour of soil conforming to ASTM D 2487 Soil Classification Chart, type SC, having a plasticity index (PI) of between 10 and 25 inclusive, to at least 95 percent Modified AASHO in compacted lifts of not less than 6 inches. The compactor shall at a minimum, be capable of negotiating a 10 percent grade over material offering a rolling resistance of 150 pounds per ton or greater, for a total effective grade of 17.5 percent. The compactor shall be capable of operating in ambient temperatures between 125 and -25 °F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperatures in the sump and gallery, and hydraulic component manufacturer's fluid temperature limits.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Production compactors. All compactors furnished under this specification shall be physically and mechanically identical to the first article compactor with modifications approved by the Government (see 6.3).

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials, coatings and special treatments utilized shall provide protection against the various forms of material deterioration in the applicable operating and storage environments to which the compactor may be exposed.

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3.4.1 Material deterioration prevention and control. The compactor shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the compactor may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the compactor may be newly fabricated from recovered materials to the maximum extent practicable, provided the compactor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the compactor.

3.5 Maintenance.

3.5.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers or plates which must be removed or opened for component adjustment or maintenance shall be equipped with quick-opening, hand-operable fasteners and shall be labeled to indicate the items accessed through them. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using special tools furnished with the compactor and common tools. After completion of any maintenance operation, the compactor shall operate as specified herein. Electrical system fuses shall be grouped at a common location. The fuses to be changed without the use of tools. Each fuse holder shall be labeled as to equipment served by the fuse and the fuses current rating (i.e., lights 20A). The label shall be readable when the fuse is in place.

3.5.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.5.3 Fluid reservoirs. Fluid reservoir fill ports, including fuel tank and radiator, shall be labeled on or near the port to indicate the type of fluid used (i.e., "DIESEL FUEL", "ENGINE OIL", "COOLANT", etc). Except for the fuel tank,

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reservoir fill ports shall be labeled on or near the port with the quantity of fluid required to fill the reservoir (i.e., "4 GALLONS", "3 QUARTS", etc.). Except for the fuel and radiator, the specific type of fluid shall be labeled on or near the fill port.

3.5.4 Filters. Except for the transmission filter, all filters shall be changeable without the removal of other compactor components. Removal of access covers to change filters shall be acceptable. The air filter shall be changeable without the use of tools.

3.6 Safety. All exposed parts which are subject to high operating temperatures, or which are energized electrically shall be insulated, fully enclosed or guarded. All rotating, reciprocating or moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. All walking surfaces shall be provided with an anti-skid surface. Unless otherwise specified (see 6.2), seat belts shall be provided and the belts shall be anchored in accordance with SAE J386. Operator visibility shall be adequate to permit safe operation in forward and reverse travel. All platforms, steps, and hand holds shall be in accordance with SAE J185 and permit unhampered, non-hazardous entrance and exit from the equipment. A reverse alarm of the audible type with a manual override shall be provided.

3.6.1 Rollover protective structure (ROPS). Unless otherwise specified (see 6.2), the compactor shall be equipped with a removable bolted down ROPS. The ROPS shall conform to SAE J1040. A slinging eye(s) shall be provided to facilitate ROPS removal.

3.7 Noise limits. Noise levels produced by the compactor shall comply with the requirements of MIL-STD-1474, except as indicated herein.

a. The noise level at the operator's position and at one meter from the perimeter of the compactor at all other locations shall not exceed Category D of table I (85 dB(A). MIL-STD-1474, 5.1.1.2, 5.2, 5.3 and 5.4 shall not apply. Compliance and non-compliance with the required noise limits shall be documented based upon test requirements of 4.5.2.6 and the data requirements of MIL-STD-1474, 5.5.
b. The following table I shall be used in lieu of table II contained in MIL-STD-1474B:

TABLE I. Steady-state noise limits (dB) for personnel-occupied areas.

	Limit category					
	A*	B*	C*	D*	E**	F**
A-weighted limit ([dB(A)]	108	100	90	85	75	65

3.8 Frame. The frame shall be of the articulated type. The frame or wheel assembly shall be constructed to allow wheel oscillation.

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3.8.1 Steering system. The compactor shall be equipped with a hydraulic powered articulated (center point) type steering system which provides for tracking of front and rear wheels or drums of full width coverage. Compactors, built in two sections with a hingepoint for articulation, shall have steering frame locks conforming to SAE J276. A caution plate shall be posted on each side of the compactor near the articulation point cautioning personnel not to stand in the area of the articulation point when the engine is running and to use the steering locks when servicing the equipment and during transport. The force required by the operator to turn the wheel from lock to lock shall not exceed 50 pounds when measured at the wheel rim with the compactor stationary and the power assist system disconnected.

3.8.2 Lighting system. The compactor shall be furnished with front and rear mounted headlamps or floodlamps for illumination of the area in front of and behind the compactor for safe and efficient night operation of the compactor.

3.9 Propulsion system. The compactor shall be equipped with a torque converter and a power shift transmission having at least 3 gear speed ratios. A single forward-reverse lever shall be provided to control the direction of the compactor for each of the 3 minimum gear speed ratios. Engine power may also be transmitted through a full hydrostatic drive with infinitely variable speed control in forward and reverse. The compactor engine starting system shall be such that the transmission must be placed in neutral position before the engine can be started.

3.10 Brake system. The compactor shall be provided with an air, an air-over-hydraulic, or a full hydraulic service brake system. Service brakes shall be capable of bringing the compactor to a complete stop within 25 feet from a speed of 10 mph on a dry level surface. In addition, a manual or automatic emergency parking brake shall be provided. The parking brake shall be capable of holding, but not necessarily stopping, the compactor on a 15 percent grade with the engine running, for a period of 15 minutes without the assistance of the service brakes, transmission lock, wheel chocks, or any other means. The force required by the operator to apply the service brake shall not exceed 37 pounds. The force required by the operator to apply the service brake with the power assist system disconnected shall not exceed 70 pounds. The force required by the operator to apply the parking brake shall not exceed 24 pounds for hand actuated parking brakes or 57 pounds for foot actuated parking brakes.

3.10.1 Brake fluid. When hydraulic brakes are utilized, the brake system shall be filled with silicone brake fluid conforming to MIL-B-46176.

3.11 Strike-off (leveling) dozer blade. The leveling blade shall be hydraulically operated and shall have a replaceable cutting edge. The blade width shall be wider than the compactor width, but shall not exceed the compactor width by more than 6 inches on each side. The blade height shall be not less than 30 inches. Hydraulic controls shall include lift, lower, float and hold positions.

3.11.1 Compaction monitoring device. The compactor shall have a comparative compaction monitoring device capable of detecting soil density during normal compaction operations. The device shall monitor relative load-bearing conditions for various soil types during compaction. At the operator's station, an analog

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or readout provisions shall continually monitor the soil and the compaction effort. The readout shall show when maximum compaction is attained. The device shall receive power from the compactor's electrical system. The sensors and transducers shall be protected during operations and cleaning. All exposed wiring shall be protected in conduit.

3.12 Wheels and cleaners. Each compactor wheel or drum shall be provided with tamping feet or segmented impact pads. The tamping foot tips or segmented impact pad caps shall be of the replaceable type, either bolt-on, pin-on or weld on. Cleaners shall be provided fore and aft, on all drums or wheels. The cleaners shall be adjustable and replaceable and shall prevent material build-up regardless of direction of travel.

3.13 Control station. The compactor shall be equipped with a operator control station. The station shall be sized in accordance with SAE J154. Access and egress from the station shall be in accordance with SAE J185.

3.13.1 Seating. The control station shall be equipped with a seat conforming to SAE J899. The seat shall be covered with a durable, waterproof covering. The seat shall be located such that small woman through large man operators, as defined in SAE J833, shall have an unobstructed field of view fore, aft, and to either side of the compactor while seated in a normal operating position.

3.13.2 Instruments. The control station shall be equipped with all instruments, gauges, and indicators necessary for operation of the compactor. As a minimum, the station shall be equipped with an ammeter or voltmeter, an oil pressure gauge, a fuel gauge, and an engine coolant temperature gauge. Instruments shall be labeled as to the measured quantity. Pictograms conforming to SAE J1362 may be used in lieu of text labels. Instruments shall be located in accordance with SAE J209.

3.13.3 Controls. The control station shall be equipped with all controls necessary for operation of the compactor. As a minimum, the station shall be equipped with light system controls, ignition switch, throttle, transmission control, service and parking brake controls, and a steering wheel. Controls shall be designed and located in accordance with SAE J297 and J898. Except for the steering wheel and service brake, all controls shall be labeled as to function and with the minimum information needed for proper actuation. Pictograms conforming to SAE J1362 may be used in lieu of text labels.

3.13.4 Security protection. Unless otherwise specified (see 6.2), the compactor shall be equipped with locking caps to secure the following: Fuel tank, hydraulic oil reservoir, engine oil filler, and radiator, unless protected by engine hood and easy access hood, side doors or panels that can be locked.

3.14 Diesel engine. The compactor shall be powered by industrial diesel engine having horsepower, torque, and speed characteristics to satisfactorily meet all performance requirements specified herein without exceeding the engine manufacturer's intermittent rated horsepower of the engine furnished. The engine governed speed shall be set and sealed in accordance with the engine manufacturer's recommendations. The engine shall be capable of operating continuously at rated load in an ambient temperature of 125 °F without exceeding the maximum allowable heat temperatures established by the engine manufacturer.

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The engine shall operate on either JP-8 turbine or diesel fuel conforming to MIL-T-83133 or VV-F-800, respectively. The engine shall be furnished with the following accessories:

a. Ammeter or voltmeter.
b. Oil pressure indicator.
c. Oil filter.
d. Air cleaner, dry type.
e. Fuel gauge.
f. High temperature cut off switch.
g. Low oil pressure cut off switch.

3.14.1 Engine fuel tank. The fuel tank shall be of sufficient capacity for not less than 8 hours operation at the maximum fuel rating of the engine. The fuel tank shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Fuel lines shall be protected against mechanical damage. Construction and mounting shall be such that neither vibration nor tightening of mounted devices will cause leaks to develop. The bottom of the fuel tank and its supporting members shall not be lower than the lowest adjacent part of the chassis or body. The fuel fill port shall be labeled "DIESEL OR JP-8 FUEL ONLY" with cast or embossed letters not less than 1 inch letter height.

3.14.2 Engine oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. The oil sampling valve shall be labeled "ENGINE OIL SAMPLE". It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running.

3.14.3 Diagnostic connector assembly (DCA). The compactor shall incorporate an easily accessible DCA in the operator's cab for interface with the Government's Simplified Test Equipment/Internal Combustion Engine (STE/ICE) in accordance with Report No. CR-82-588-003 and MIL-T-62314. As a minimum, the DCA shall be capable of monitoring the functions defined in table II of Report No. CR-82-588-003 and alternator/generator output voltage, alternator/generator negative cable voltage drop, fuel solenoid voltage, starter voltage, starter current first peak and average, starter circuit resistance, battery internal resistance change, fuel supply pressure, and fuel filter restriction (Delta P). Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Fittings/adapters required for the interface with the STE/ICE test set when monitoring in the TK mode shall be permanently installed on the compactor. Adaptors in the STE/ICE test set shall readily mate with test point connections in the compactor. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. The contractor shall also provide STE/ICE vehicle test cards in the format identified in Drawing No. 12258955 addressing both DCA and TK measurements.

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3.15 Electrical system. The compactor shall be furnished with a 12 volt or 24 volt electrical cranking system including the following components:

a. Batteries.
b. Starter.
c. Alternator.
d. Voltage regulator.
e. Ignition switch.
f. Slave receptacle.

3.15.1 Battery compartment and cables. The battery storage compartment shall be designed to allow the small woman through the large man, as defined in SAE J833, to replace the batteries and to check and fill the acid level of non-sealed batteries. A pictorial battery connection diagram shall be provided. The diagram shall be viewable during battery maintenance. Each battery terminal shall be protected by an insulated boot. The boots shall be labeled "+", colored red for the positive terminal and "-", colored black for the negative terminal.

3.16 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptibility characteristics of the compactor shall conform to SAE J551, devices category.

3.17 Bracket assembly, rifle stowage. A rifle stowage bracket assembly conforming to Drawing 13229E8014 shall be furnished and mounted by the contractor. The location of the bracket shall be such that it is accessible to the operator and provides clearance for the rifle to be removed. The bracket shall be located away from operator's controls and shall not impair the vision of the operator.

3.18 Transportability. The compactor shall be capable of being transported world wide by highway, rail, marine, and air moded in C-130, C-141, and C-5A aircraft. The applicable air transportability criterion is as set forth in MIL-STD-1791. The compactor shall withstand impact forces encountered in rail and air shipments without damage or permanent deformation. The compactor shall not exceed the Gabrit Internation De Chargement (GIC) clearance dimensions shown in figure 1 for rail transport. The vehicle slinging and tiedown provisions shall withstand all forces to be encountered during loading or transportation. During preparation for internal air transport, the exhaust stack and ROPS may be relocated to meet the air transportability requirements. Reassembly after transport in C-130 and C-141 aircraft shall be accomplished in 60 minutes or less using only a mechanic with hand tools and a lifting device with operator. The compactor shall have 25 percent of its fuel supply and shall be fully serviced.

3.18.1 Rail impact. The compactor shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.5.2.10 without failure, damage, or permanent deformation.

3.18.2 Marine transportability. The compactor shall be transportable by ships in accordancw with MIL-HDBK-157.

3.18.3 Lifting provisions. The lifting provisions shall conform to MIL-STD-209, class 1 or 3 provisions. The provisions shall enable the complete compactor to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between lifts and

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all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural component. Lifting provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.18.4. All lifting/tiedown provisions shall be labeled "LIFT", "TIEDOWN", or "LIFT/TIEDOWN", as appropriate, in 1-inch (2.54 cm) high letters.

3.18.4 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3 equipment. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.19 Towing attachments. Towing attachment(s) shall be provided. The attachment(s) shall be hook(s) or loop(s) suitable for towing (or recovering) the compactor. Instructions for the use of the attachments shall be included on plates made of material specified in 3.24. The instruction plates shall be permanently affixed to the compactor.

3.20 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. A lubrication chart in accordance with SAE J753 shall be furnished with each compactor.

3.20.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.20.2 Pressure-release device. An automatic pressure-release device shall be provided where pressure-lubricating equipment could damage grease seals or other parts.

3.20.3 Filling, draining, and checking provisions. Enclosures such as gearcases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipstick, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure into a suitable container. Integral tubes or troughs may be used to satisfy this requirement.

3.20.4 Grease lubrication. All grease lubrication shall be with greases conforming to MIL-G-10924 or other military greases, as applicable. Information on MIL-G-10924 and other military greases may be found in MIL-HDBK-113. The compactor shall be assembled, run-in, tested, and delivered with these greases.

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The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.20.5 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with winter grades of MIL-L-2104 lubricating oil, military symbol OE/HDO-10 or OE/HDO-15/40 designated for use in the temperature range of 0 to 32 °F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.20.6 Servicing and adjustments. The contractor shall service and adjust each compactor for correct operation and travel including at least the following: adjustment of the engine; adjustment of the brake system and electrical system; complete lubrication of the compactor with the types and grades of lubricant recommended.

3.21 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.21.1

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.21.1 Composition. Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air-Quality Management District.

3.22 Shipping data plates. Shipping data plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and shall show the silhouette of the compactor in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown attachments. The recommended lifting and tiedown arrangement including spreader bar arrangement shall be included on these plates. The data plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.23 Identification marking. The compactor shall be identified in accordance with MIL-STD-130. The markings shall be applied to the compactor on a plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.24 Control identification, safety and warning plates. A plate shall be furnished for all points requiring attention for safe and efficient operation of the compactor. The plates shall be labeled by lettering or other appropriate marking with information such as neutral position of each control, safety instructions, and other warnings or instructions, as necessary. The plates shall be anodized aluminum, sunken-etched or raised relief. Plates that describe safety measures or that warn against operations detrimental to the compactor shall be distinguished by the words "CAUTION" or "WARNING" in red lettering. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.25 Treatment and painting. The portions of the compactor normally painted, internally and external, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), the top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2), the compactor shall be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383.

3.25.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.26 Stenciling. The gross weight of each compactor shall be stencilled on each side of the compactor. All stenciling shall be in block-type letters not less than 1 inch high, and shall be black in color.

3.27 Tool box. The compactor shall be provided with a closed compartment or tool box to carry special tools and accessories required for operation and operator maintenance of the equipment.

3.28 Special tools. Special tools required for maintenance and repair, (if required) shall be supplied by the manufacturer for each item of equipment upon delivery and shall be packed with the equipment. Special tools shall be kept to a minimum.

3.29 Workmanship.

3.29.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.29.2 Bolted connections. All bolt holes shall be accurate punched or drilled and shall have the burrs removed. Washers or lock-washers shall be provided. All nuts, bolts, and screws shall be tight.

3.29.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet heads shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

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3.29.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code such as AWS D14.3. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.29.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defects which reduces the castings ability to perform its intended function.

3.30 Government furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.7).

Item No.	Description	Quantity
1	Camouflage paint pattern.	As required
2	Cases complete with maintenance forms Case NSN 7520-00-559-9618.	As required
3	Vehicle Weight Classification Sign Kit NSN 9905-00-565-6267.	As required
4	Report CR 82-588-003 STE/ICE Design Guide for Vehicle Diagnostic Connector.	As required

3.31 Decontaminating apparatus, portable (DAP), M13. The compactor DAP mounting bracket (MS53052-1) shall be installed in a manner which shall not interfere with the normal operation of the equipment. An upright installation is preferred. Horizontal mounting shall be used only with the approval of the contracting officer. The DAP device has a flash point of 165 °F and shall be installed away from exhaust or heat sources.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The compactor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The compactor shall be tested as specified in 4.5.2.1 through 4.5.2.13. All first article tests shall be run using JP-8 fuel conforming to MIL-T-83133. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each compactor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each compactor shall be completely assembled, properly adjusted, and otherwise serviced for operation. Each compactor shall be operated for not less than 2 hours. The compactor shall be driven in both the forward and reverse mode for a total distance of not less than 5 miles. All controls shall be operated as many times as necessary, but not less than 30 times, to determine ease of operation, effectiveness, and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. Gearcases, bearings, and other moving parts shall be examined for excessive heating or abnormal operation insofar as is practicable and possible. Tests shall be run using JP-8 or diesel fuel conforming to MIL-T-83133 and VV-F-800, respectively (see 3.14). The entire compactor shall be examined for evidence of failure, slippage, or deformation. Failure to pass any phase of this test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. The compactor shall be examined for the following defects:

101. Compactor not assembled as specified (see 3.1).
102. Dimensions not as specified (see 3.1.1).
103. Materials not as specified (see 3.4).
104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.4.1).
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.4.2).
106. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.4.3).

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107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the compactor (see 3.4.4).
108. Maintainability not as specified (see 3.5.1).
109. Interchangeability not as specified (see 3.5.2).
110. Fluid reservoirs not labeled as specified (see 3.5.3).
111. Filters are not changeable without the removal of other components (see 3.5.4).
112. Safety devices missing or not as specified (see 3.6).
113. Rollover protective structure (ROPS) not as specified (see 3.6.1).
114. Frame not as specified (see 3.8).
115. Steering system or locking device not as specified (see 3.8.1).
116. Lighting system not as specified (see 3.8.2)
117. Propulsion system not as specified (see 3.9).
118. Brake system not as specified (see 3.10).
119. Brake fluid not as specified (see 3.10.1).
120. Dozer blade not as specified (see 3.11).
121. Wheels and cleaners not as specified (see 3.12).
122. Control station not as specified (see 3.13 thru 3.13.3).
123. Security protection not as specified (see 3.13.4)
124. Engine and accessories not as specified (see 3.14).
125. Engine fuel tank not as specified (see 3.14.1).
126. Engine oil sampling valve not as specified (see 3.14.2).
127. Diagnostic connector not as specified (see 3.14.3).
128. Electrical system not as specified (see 3.15).
129. Battery compartment and cables not as specified (see 3.15.1).
130. Electromagnetic interference not as specified (see 3.16).
131. Rifle stowage bracket assembly not as specified (see 3.17).
132. Transportability not as specified (see 3.18).
133. Marine transportability not as specified (see 3.18.2).
134. Lifting and tiedown attachments not as specified (see 3.18.3 and 3.18.4).
135. Towing attachments not as specified (see 3.19).
136. Lubrication system not as specified (see 3.20 thru 3.20.3).
137. Grease lubrication not as specified (see 3.20.4).
138. Oil lubrication not as specified (see 3.20.5).
139. Servicing and adjustments not as specified (see 3.20.6).
140. Fungus and moisture resistance not as specified (see 3.21 and 3.21.1).
141. Shipping data plates missing or not as specified (see 3.22).
142. Identification marking not as specified (see 3.23).
143. Control identification, safety and warning plates not as specified (see 3.24).
144. Treatment and painting not as specified (see 3.25).
145. Camouflage pattern data not provided as specified (see 3.25.1).
146. Stenciling not as specified (see 3.26).
147. Tool box not as specified (see 3.27).
148. Special tools not as specified (see 3.28).
149. Workmanship not as specified (see 3.29 thru 3.29.5).
150. Decontaminating mounting bracket not as specified (see 3.31).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The compactor shall operate as specified herein without maintenance other

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than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Performance. Operate the compactor on a fill, built over a natural soil subgrade having an initial density of at least 90 percent Modified AASHO. The fill material shall correspond to group SC, having a plasticity index of 10-25, of the Soil Classification Chart in ASTM D 2487. The compacted fill site shall be not less than 30 feet wide, with the actual width of the fill being a minimum of 3 feet wider than the compacted area. The fill may be placed prior to the start of compaction operation, during compaction operations, or the contractor may elect to till the natural soil in lieu of placing fill material. When hauling equipment is run over spread, uncompacted fill material, the fill site shall be sacrificed or tilled to a uniformly loose uncompacted condition. Spreading of the fill material shall be done with a grader. Spreading runs shall be only that necessary to reach the estimated uncompacted fill lift thickness. Water distribution equipment may be used as required. The material to be compacted shall be of sufficient thickness, initially, to give a compacted lift of at least 6-inches. The laboratory density of the soil to be compacted shall be determined in accordance with ASTM D 1557, method optional. When the contractor feels that a compaction rate of not less than 1500 cubic yards of fill to a density of 95 percent modified AASHO has been accomplished, he may stop compaction operations. Time used in density testing shall not be counted as compaction time. Density measurements shall be taken at points along the test strip using the pattern shown on figure 2 at intervals of 200 feet. In addition, a minimum of 3 samples shall be taken at intervals of 100 feet and at random locations specified by the Government. Density testing shall be accomplished by a nuclear density meter. The nuclear density tester shall have been calibrated against the soil to be compacted. An achieved fill material density of less than 95 percent Modified AASHO or less than 6-inches in depth shall constitute failure of this test. Sufficient measurements shall be taken to determine the in-place quantity of material compacted. The above procedure shall be repeated for a total of 3 times. The average compacted cubic yards shall be determined.

4.5.2.2.1 Failure Criteria. An average compaction rate of less than the compacted cubic yards per hour specified in 3.1.2 and failure to achieve compacted lift thickness of at least 6 inches for each test shall constitute failure of this test.

4.5.2.3 Gradeability. Operate the compactor on a minimum grade of 10 percent over fill material having a rolling resistance of not less than 150 pounds per ton (total effective grade of 17.5 percent). The offeror may, at his option, operate the compactor up a grade of at least 17.5 percent.

4.5.2.3.1 Failure criteria. Inability to negotiate the grade under either condition specified above shall constitute failure of this test.

4.5.2.4 High temperature operation. The contractor shall provide proof (test data or calculation) that the compactor can be operated in ambient temperatures up to and including 125 °F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperature in the sump and gallery, and hydraulic component manufacturer's fluid temperature limit.

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4.5.2.4.1 Failure criteria. Failure to meet this requirement shall constitute failure and acceptance of the equipment shall not be approved until corrective action has been taken and approved by the Government.

4.5.2.5 Electromagnetic interference. Unless otherwise specified, the compactor shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article test report.

4.5.2.5.1 Failure criteria. Failure to meet the requirements of SAE J551, shall constitute failure of this test.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level.

4.5.2.6.1 Failure criteria. Failure to comply with MIL-STD-1474 or 3.7 provisions shall constitute failure of this test.

4.5.2.7 Brake test. The compactor shall be completely stopped five times as specified in 3.10 within a period of 10 minutes, to test brake fadeout. The distance from point of brake application to actual stop shall be measured on the fifth pass to determine conformance of 3.10. The service brake test shall be conducted in both forward and reverse. With the roller stopped on a 15 percent slope, the parking brakes shall be applied and shall be capable of securing the compactor in the stopped position without creeping. The brake test shall be conducted with the compactor heading up and down slope.

4.5.2.7.1 Failure criteria. Failure of the compactor to meet the braking requirements specified in 3.10 shall constitute failure of this test.

4.5.2.8 Rail impact test. The compactor shall be tested in accordance with MIL-STD-810 to prove conformance to 3.18.1. The rail cars shall be equipped with standard friction draft gears as defined by the Association of Americal Railroads (AAR) rules (without coupler cushioning devices) and conventional underframes.

4.5.2.8.1 Failure criteria. Any permanent deformation, failure, or inoperability after test shall constitute failure of this test.

4.5.2.9 Lifting provisions test. The lifting provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.18.3.

4.5.2.9.1 Failure criteria. Inability to meet the requirements of 3.18.3 shall constitute failure of this test.

4.5.2.10 Tiedown provisions test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.18.4.

4.5.2.10.1 Failure criteria. Inability to meet the requirements of 3.18.4 shall constitute failure of this test.

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4.5.2.11 Compaction monitoring device. The compactor shall be operated over a an area of fill selected by the contractor until the device indicates that maximum compaction has been reached. The compacted area shall be tested using a nuclear density device to determine maximum density. The compactor shall then be operated until the monitoring device indicates compaction has decreased. The nuclear density device shall be used to determine actual density.

4.5.2.11.1 Failure criteria. If the monitoring device indicates a lower reading and the density of the fill material has increased or nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.12 DCA capability. The contractor shall perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the compactor to ensure wire and connections are free of error. After installation on the compactor, test the DCA harness (see Report No. CR-82-588-003) on each vehicle with a DCA tester prior to interfacing with STE/ICE.

4.5.2.12.1 Failure criteria. Nonconformance to 3.14.3 shall constitute failure of this test.

4.5.2.13 DCA-STE/ICE interface. Upon completion of inspection and tests specified in 4.5.2.12, test the DCA for all measurements specified in 3.14.3 using the STE/ICE.

4.5.2.13.1 Failure criteria. Nonconformance to 3.14.3 or failure of any component or wiring in the DCA harness which results in inability to obtain the measurements previously recorded with the STE/ICE shall constitute failure of this test.

4.6 Inspection comparison. The Government may select compactors at any time during the contract production period and subject these compactors to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government. on units selected at random from those which have been accepted by the Government and will not include the previously inspection first article model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of these tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison compactor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison compactor and shall be cause for refusal by the Government to continue acceptance of production compactors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compactors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compactors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of packaging. The preservation, packing and marking shall be examined in accordance with the quality assurance provisions of MIL-R-3075 and as specified in section 5 herein.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A or B. Each complete compactor shall be preserved as specified for level A or B requirements for rollers, in accordance with MIL-R-3075.

5.1.2 Level C. Each complete compactor shall be preserved in a manner to afford protection against deterioration and damage during shipment from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A or B. Each complete compactor shall be packed as specified by the level A or B requirements for rollers, in accordance with MIL-R-3075.

5.2.2 Level C. Each complete compactor shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. The compactor shall be marked as required for rollers in accordance with MIL-R-3075.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The high speed, segmented compactor covered by this specification will be used for embankment compaction of SC type soils during the construction of roads, airfields, and dams.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required (see 3.2).
d. Time frame for submission of first article and number of compactors required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests (see 3.2). When the Government will conduct some but not all of the first article examination and tests the contracting officer shall specify which examination and test will be conducted by the Government and which examination and test will be conducted by the contractor (see 3.2).
f. When seat belts are not required (see 3.6).

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g. When a rollover protective structure (ROPS) is not required (see 3.6.1).
h. When security protection is not required (see 3.13.4).
i. When electromagnetic interference requirements are other than as specified (see 3.16).
j. When fungus and moisture resistance is not required (see 3.21).
k. Top coat color when other than specified (see 3.25).
l. When overcoating with camouflage patterns is required (see 3.25).
m. When camouflage patterns will be provided (see 3.25.1 and 6.2.1).
n. Level of preservation and packing required (see 5.1 and 5.2).

6.2.1 Consideration of data requirements. The following data requirements must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
a. 3.25.1	DI-MISC-80192	Color photograph prints	
b. 3.25.1	DI-MISC-80176	Camouflage line art data	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compactors.

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6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the compactors in accordance with this specification.

6.7 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.30.

6.8 Subject term (key word) listing.

Diesel engine
Embankment
Tamping foot

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

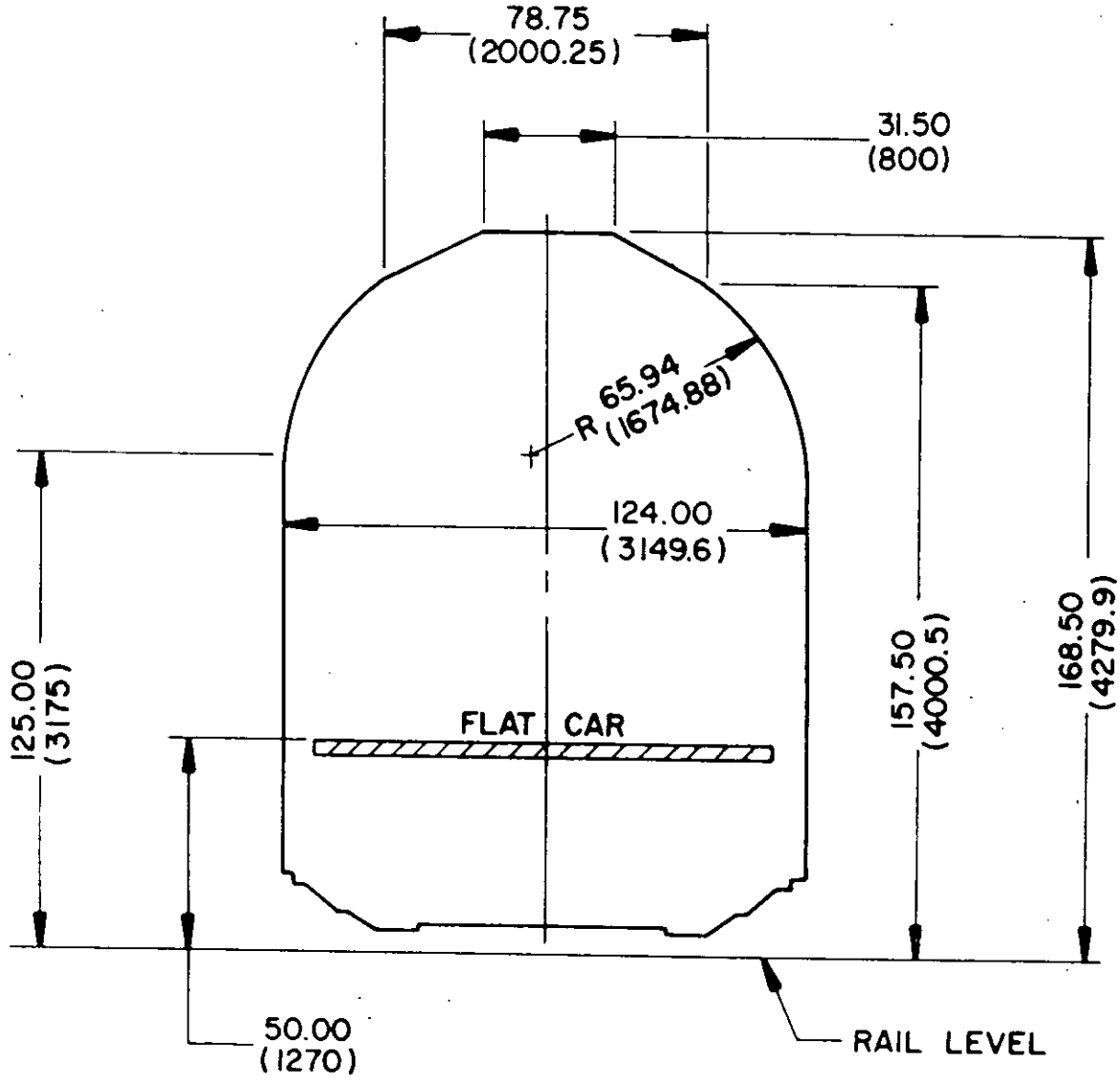
Custodian:
Army - ME

Review activity:
Army - ME

Preparing activity:
Army - ME

Project 3895-A314

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NOTE: DIMENSIONS SHOWN: INCHES (MILLIMETERS)

FIGURE 1. Gabarit International De Chargement (GIC) clearance diagram.

X-4093C

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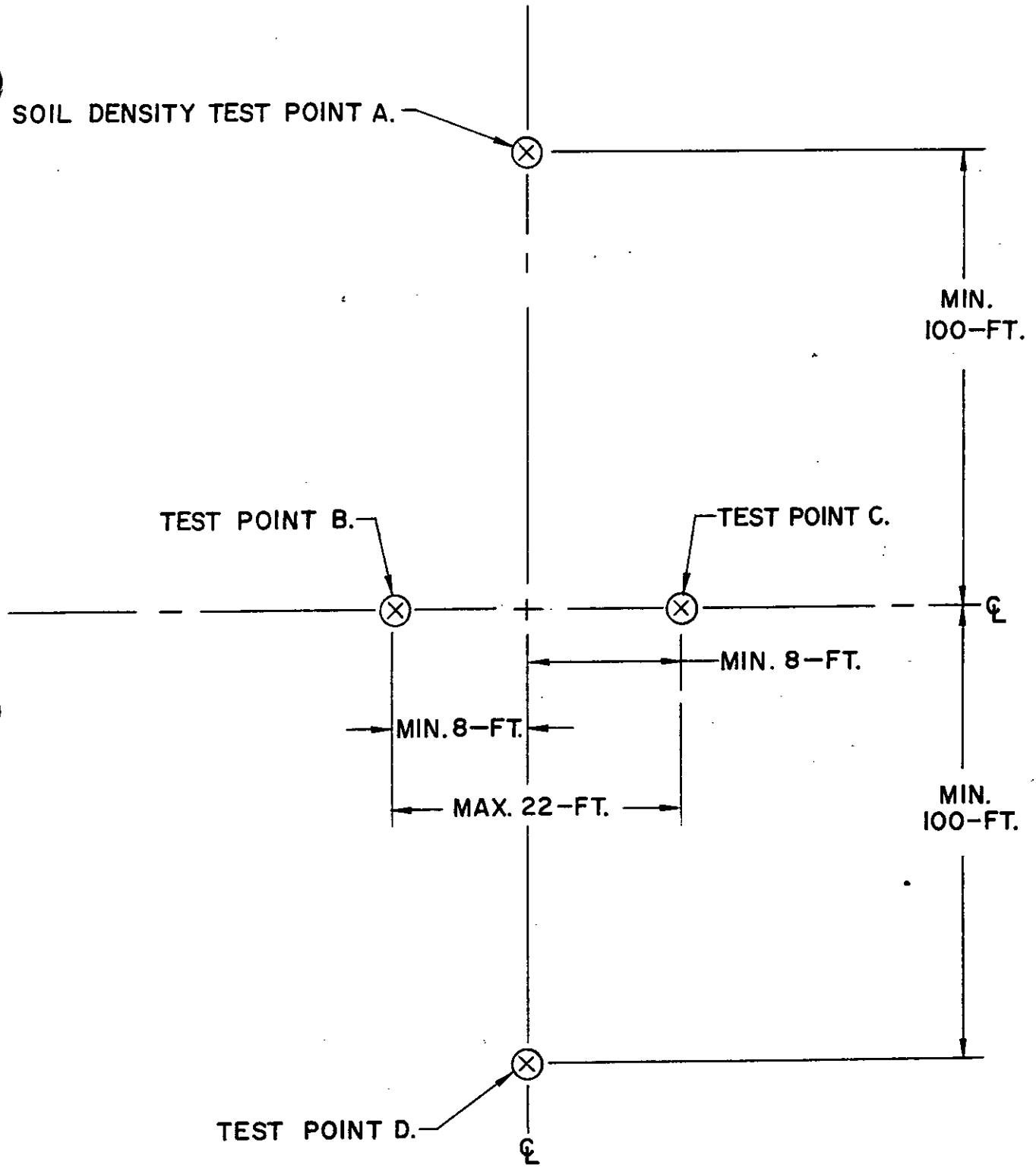


FIGURE 2. Test strip pattern.

X-4080

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-53024A(ME)	910123

3. DOCUMENT TITLE
Compactor, High Speed, Self-Propelled, Commercial Construction Equipment (CCE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-5717
	(2) AUTOVON 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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